FAZEEL ASGHAR

Lead AI Engineer
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SUMMARY

Lead AI Engineer with 5+ years in applied machine learning, including freelance work, the last two taking LLM products from design to production: low-latency voice agents, RAG platforms and tool-using agent workflows. Leads AI engineering at 7 Kings Code, owns architecture and delivery, and directs a 10-engineer team. Treats evaluation, tracing and latency monitoring as release gates. Earlier work in computer vision and IoT for healthcare; has delivered under HIPAA/GDPR and US/UK telephony consent rules.

CORE COMPETENCIES

Generative AI | Large Language Models (LLMs) | Retrieval-Augmented Generation (RAG) | Agentic AI | Conversational AI | Voice AI | Natural Language Processing (NLP) | Computer Vision | MLOps

Hybrid Retrieval and Reranking | LLM Evaluation and Observability | Model Serving | Cloud Deployment (AWS, Azure) | Real-time Telephony (SIP/VoIP)

Computer Vision: object detection and classification (YOLOv8), image segmentation, gaze estimation with calibration, OpenCV pipelines, edge inference on Raspberry Pi, model evaluation (mAP, precision/recall, F1)

Technical Leadership: system architecture, design and code review, delivery planning, model and vendor evaluation, client-facing technical discussions, cross-team coordination

PROFESSIONAL EXPERIENCE

Lead AI Engineer | 7 Kings Code LLC, Lahore, Pakistan *Sep 2025 - Present*

- Architected and led delivery of a real-time AI sales calling agent on Gemini Realtime over SIP/VoIP: 50+ languages, 100+ calls a day, 10+ CRM integrations, 7 VoIP providers, barge-in handling and p95 response latency of 200 to 300 ms. Sold as tiered plans to 10 clients. **aisaleagent.com**
- Defined the call design: the agent classifies each lead's industry, adapts the pitch, runs the full conversation and writes the transcript and disposition back to the CRM. Campaigns and pitch scripts are configurable per client.
- Drove the compliance work (AI disclosure, opt-out, Do-Not-Call suppression, recording consent) so one product runs under TCPA (US) and PECR/GDPR (UK) rules.
- Led the build of an AI recruitment calling agent connected to 5+ CRMs and several ATS platforms. It pulls resumes, matches them to job descriptions, runs screening and light technical phone interviews, and moves shortlisted candidates to the next stage. Cut screening time by 40% across a 500-candidate batch. **airecruitmentagent.com**
- Directed 10 engineers on Jobsshopper, a job portal with CV parsing, JD generation, CV-to-JD matching, automated shortlisting and AI phone interviews. It parses 1,000 CVs in under 5 minutes. **jobsshopper.com**
- Own the RAG platform over a continuously changing knowledge base of 150+ documents (Pinecone, Chroma). Led the move to hybrid dense and sparse retrieval with reranking, which improved answer accuracy by 70 to 80%. Retrieval and answer quality are tracked in LangSmith, so a corpus change that hurts hit rate or faithfulness is caught before release.
- Led model training and calibration for a gaze-estimation system built for an international advertisement-testing research team, and tuned it to their testing requirements. **cortality.com**
- Own tracing, latency monitoring and releases for the voice products. Run technical planning, design reviews and code reviews for the team.

AI Engineer | 7 Kings Code LLC, Lahore, Pakistan *May 2025 - Sep 2025*

- Delivered a multi-session conversational agent with session memory for the Unitree Go2 Pro quadruped robot, covering interaction and task execution.
- Built the GenAI resume parser behind the Jobsshopper platform, an email-triage agent that classifies incoming messages and alerts the support team, and a multilingual support agent that opens Freshdesk tickets through the API.
- Developed TrueDrive, an internal Google Drive-style file platform, on Django REST Framework.

ML Engineer (Full-time) | Grant Manpower Ltd, Lahore, Pakistan *Sep 2024 - May 2025*

- Owned design and delivery of an inbound technical-support voice agent in English, German and Spanish. It took 20 to 30 live calls a day and created support tickets automatically. 50% of queries were resolved without a human, at 5 to 10 s average response latency.
- Extended the same multilingual support and ticketing workflow to a text chatbot.

Research Associate, AI Engineer | University of the West of Scotland, Paisley, UK *Jul 2024 - Aug 2024*

- Co-developed an AI smart wheelchair (Safe-RH sub-initiative) with IoT vital-sign monitoring. YOLOv8n Lite obstacle detection ran on a Raspberry Pi with an Arduino UNO and ultrasonic sensors and reached 95% mAP.
- Built the pipeline that streams patient vitals from the IoT sensors to a remote MySQL database for live review by care staff.

Junior AI Engineer (Full-time) | Safe-RH, Bahawalpur, Pakistan *Aug 2022 - May 2024*

- Worked on a rural health-monitoring system for areas with limited clinic access. IoT medical devices and Raspberry Pi units fed a Django platform so doctors and paramedics could watch patient vitals remotely. **safe-rh-mis.com**
- Trained regression and classification models to flag heart disease and critical conditions from live patient data.
- Built a YOLOv8 model that detects and classifies urine-sediment particles to automate microscopy analysis (90% mAP, 10,000 images).

TECHNICAL SKILLS

LLMs and agents: LangChain, LangGraph, LlamaIndex, tool/function calling, stateful multi-step agents, structured output validation (Pydantic), guardrails, prompt versioning, LoRA/QLoRA fine-tuning, Hugging Face Transformers, text generation and classification, Stable Diffusion, n8n

RAG and vector search: Pinecone, Chroma, FAISS, Qdrant, Weaviate; chunking strategies (recursive, parent document), dense + sparse hybrid retrieval, metadata filtering, query rewriting, cross-encoder reranking, multilingual embeddings, incremental re-indexing, grounded generation with citations

Model serving: vLLM, Ollama (local and on-premise inference), AWS Bedrock, SageMaker, inference APIs

Evaluation and observability: RAGAS, LangSmith, Langfuse, Arize Phoenix, MLflow, Prometheus, Grafana; retrieval metrics (hit rate@k, MRR, nDCG, context precision/recall), generation metrics (faithfulness, answer relevancy), LLM-as-judge with human spot checks, regression suites; production metrics (p50/p95 latency, resolution and handoff rate, WER, token cost per call)

Voice and real-time: Gemini Realtime API, OpenAI Realtime API, Whisper (STT), Kokoro (TTS), SIP/VoIP, Twilio, Asterisk, FreeSWITCH, WebRTC, WebSockets, barge-in handling, multilingual call flows

Data and orchestration: Databricks, Apache Spark (PySpark), Apache Kafka (working knowledge); Airflow, Prefect, Redis, Celery; PostgreSQL, MySQL, MongoDB, SQL

MLOps and cloud: Docker, Kubernetes, GitHub Actions CI/CD, MLflow, DVC; AWS (EC2, ECS, ECR, Lambda, S3, IAM, VPC, Bedrock, SageMaker); Azure (Virtual Machines and core services)

Backend and testing: Python (primary), FastAPI, Flask, Django REST Framework, REST APIs, Pydantic, Pytest, microservice design, Bash, Git, Linux

Machine learning: PyTorch, TensorFlow, scikit-learn, neural networks, regression and classification, feature engineering, predictive modelling, model evaluation

Edge and robotics: Raspberry Pi, Arduino, IoT sensor integration, embedded computer vision

SELECTED PROJECTS

- Medical Consultation Voice Agent (Dec 2024). Designed a RAG pipeline over proprietary healthcare data with Whisper STT and Kokoro TTS running locally, so no audio leaves the machine. github.com/Fazeel-AIML/Voice_Medical_Chatbot
- Agentic Travel Booking Assistant (Mar 2024). LangGraph agent that recommends destinations and books flights through multi-step reasoning and API calls. Private client.
- MLOps microservices. Vehicle insurance prediction and weather forecasting services, each containerised with Docker and shipped through GitHub Actions CI/CD with MLflow and DVC. github.com/Fazeel-AIML/Vehicle_Insurance_MLOps_Deployed
- Small autonomous car (Final Year Project). Obstacle detection, path planning and GPS navigation, with a Django/PostgreSQL interface and an AWS Lambda backend. github.com/Fazeel-AIML/Autonomous_Small_Prototype
- Open source contributions. kaggle.com/fazeelasghar huggingface.co/fasghar786

EDUCATION

BS Information Technology | Islamia University Bahawalpur, Pakistan *CGPA 3.80 / 4.00*

CERTIFICATIONS

Machine Learning Specialization, DeepLearning.AI; AI Engineering Professional, Packt; Machine Learning with Python, IBM

References: Dr. Naeem Ramzan, University of the West of Scotland. Contact details available on request.